

Question 1 In which of the following cases could you use a **counter-example** as your answer?

Select All Correct Answers:

- (a) Claim: If an integer is divisible by 3, it is odd. ✓
 - (b) Claim: If $2|a$, then integer a is even.
 - (c) Claim: All Tufts undergrads spend 4 years at the university. ✓
 - (d) Claim: If $a|c$ and $b|c$, then $(a + b)|c$. ✓
 - (e) Claim: France is a part of the continent Europe.
 - (f) Claim: There exists a rat that can speak English.
 - (g) Claim: There exists a doughnut with sprinkles on it.
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Question 2 Which of the following options are a counterexample to the statement. All movies that won the Oscar Award for Best Picture are in English.

Multiple Choice:

- (a) Jojo Rabbit
 - (b) Parasite ✓
 - (c) The Office
 - (d) The Shape of Water
 - (e) American Psycho
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Question 3 Which of the following is a valid counterexample to the claim: There exists no number that is divisible by the sum of its digits.

Select All Correct Answers:

- (a) Single digit numbers $\{1,2,3,\dots,9\}$ ✓
- (b) All multiples of 5

(c) *All double-digit multiples of 10* $\{10, 20, 30, \dots, 90\}$ ✓

(d) 14

(e) *All multiples of 3* $\{3, 6, 9, 12, \dots\}$
